

Data Warehousing and Integration in Business

What is a Data Warehouse?

- Defined in many different ways, but not rigorously.
 - **A decision support database** that is maintained **separately** from the organization's operational database
 - Support **information processing** by providing a solid platform of consolidated, **historical data for analysis**.
- "A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process."—W. H. Inmon
- Data warehousing:
 - The process of constructing and using data warehouses

Data Warehouse—Subject-Oriented

- Organized around major subjects, such as customer, product, sales
- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process

Data Warehouse—Integrated

- Constructed by integrating multiple, heterogeneous data sources
 - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
 - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
 - E.g., Hotel price: currency (dollars or euros)
 - When data is moved to the warehouse, it is converted.

Data Warehouse—Time Variant

- The time horizon for the data warehouse is significantly longer than that of operational systems
 - **Operational database:** current value data
 - **Data warehouse data:** provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
 - Contains an element of time, explicitly or implicitly
 - But the key of operational data may or may not contain “time element”

Data Warehouse—Nonvolatile

- A **physically separate store** of data transformed from the operational environment
- Operational **update of data does not occur** in the data warehouse environment
 - Does not require transaction processing, recovery, and concurrency control mechanisms
 - Requires only two operations in data accessing:
 - *initial loading of data* and *access of data*

Online transaction processing (OLTP) vs. Online analytical processing (OLAP)

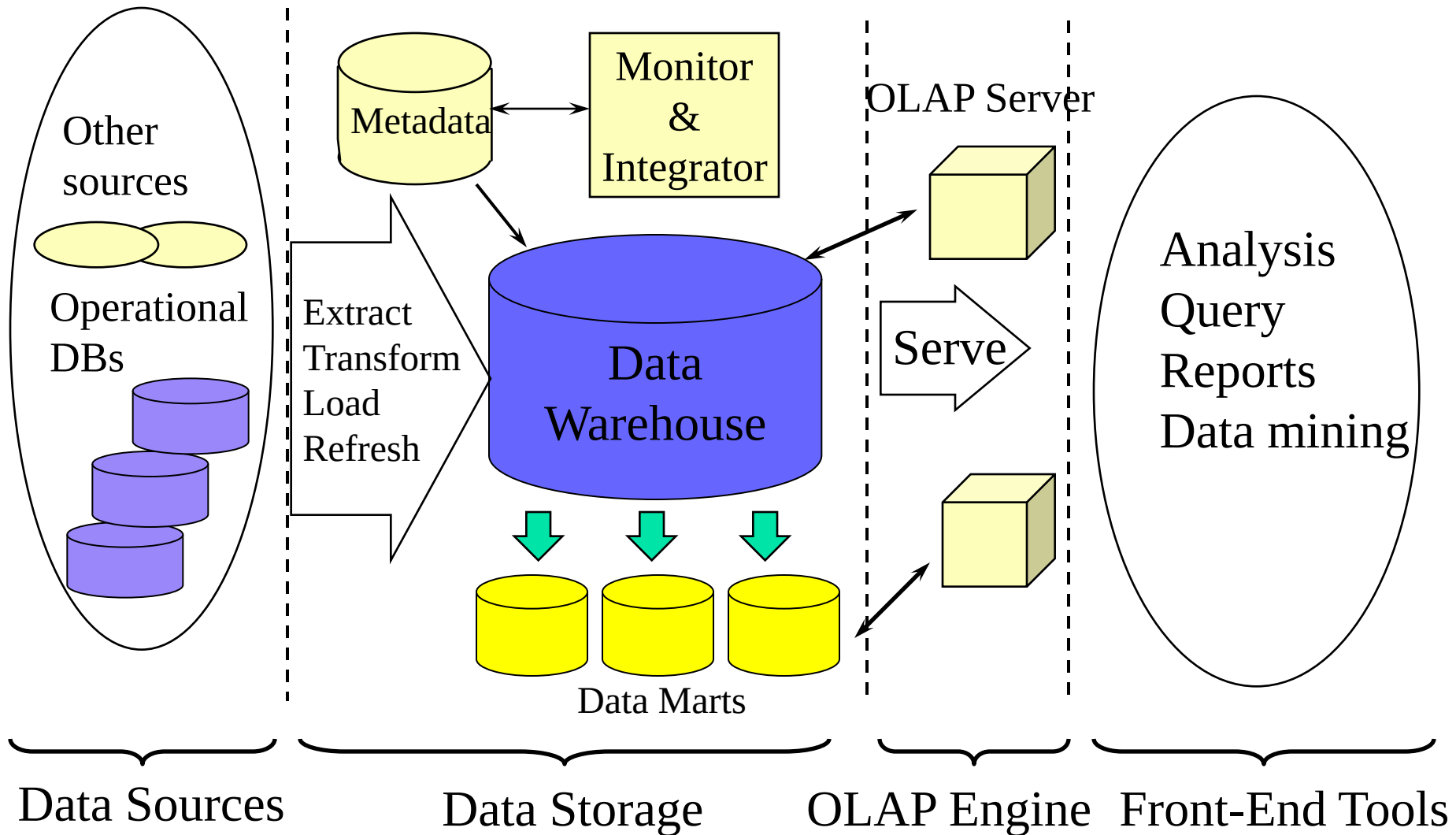
OLAP is optimized for complex data analysis and reporting, while OLTP is optimized for transactional processing and real-time updates.

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

Why a Separate Data Warehouse?

- High performance for both systems, but for different reasons:
 - **DBMS**—tuned for **OLTP**: access methods, indexing, concurrency control, recovery
 - **Warehouse**—tuned for **OLAP**: complex OLAP queries, multidimensional view, consolidation
- **Different functions and different data serve different purposes:**
 - **missing data**: Decision support requires historical data which operational DBs do not typically maintain
 - **data consolidation**: DS requires consolidation (aggregation, summarization) of data from heterogeneous sources
 - **data quality**: different sources typically use inconsistent data representations, codes and formats which have to be reconciled
- Note: There are more and more systems which perform OLAP analysis directly on relational databases

Data Warehouse: A Multi-Tiered Architecture



Three Data Warehouse Models

- **Enterprise warehouse**

- collects all of the information about subjects spanning the entire organization

- **Data Mart**

- a subset of corporate-wide data that is of value to a specific groups of users. Its scope is confined to specific, selected groups, such as marketing data mart

- **Virtual warehouse**

- A set of **views** over operational databases
- Only some of the possible summary views may be materialized

Extraction, Transformation, and Loading (ETL)

- **Data extraction**

- get data from multiple, heterogeneous, and external sources

- **Data cleaning**

- detect errors in the data and rectify them when possible

- **Data transformation**

- convert data from host format to warehouse format

- **Load**

- moves the cleaned and transformed data into the data warehouse.
 - sort, summarize, consolidate, compute views, check integrity, and build indices and partitions

- **Refresh**

- propagate the updates from the data sources to the warehouse

Metadata Repository

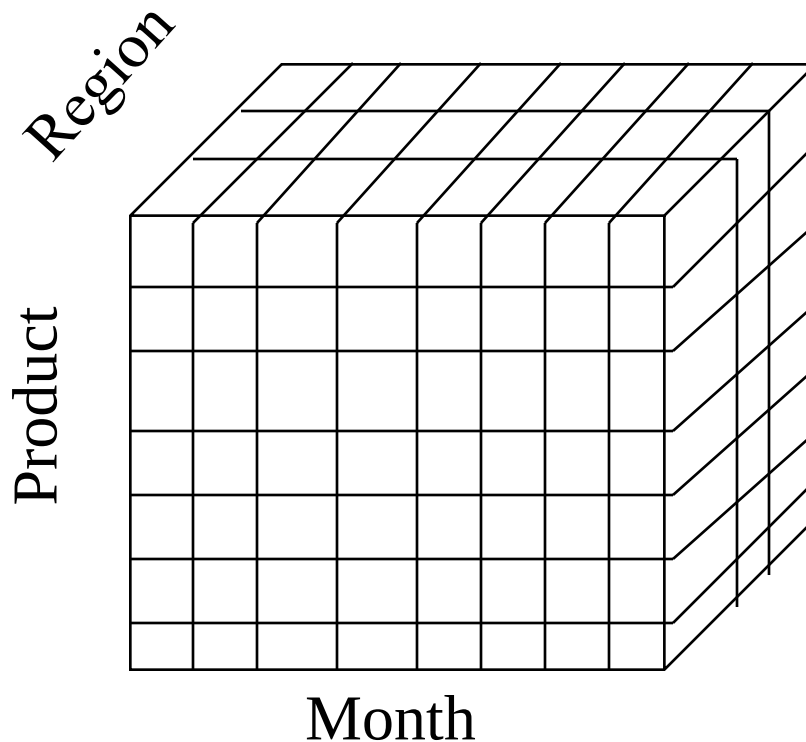
- **Meta data** is the data defining warehouse objects “data about data” . It stores:
 - Description of the **structure** of the data warehouse
 - schema, view, dimensions, hierarchies, derived data definition, data mart locations and contents
 - **Operational** meta-data
 - data lineage (history of migrated data and transformation path), currency of data (customer is “active”, “archived), monitoring information (warehouse usage statistics, error reports)
- Describes the **algorithms** used for summarization, the **mapping** from operational environment to the data warehouse and data related to **system performance**.
- Describes data related to **Business data**
 - business terms and definitions, ownership of data

From Tables and Spreadsheets to Data Cubes

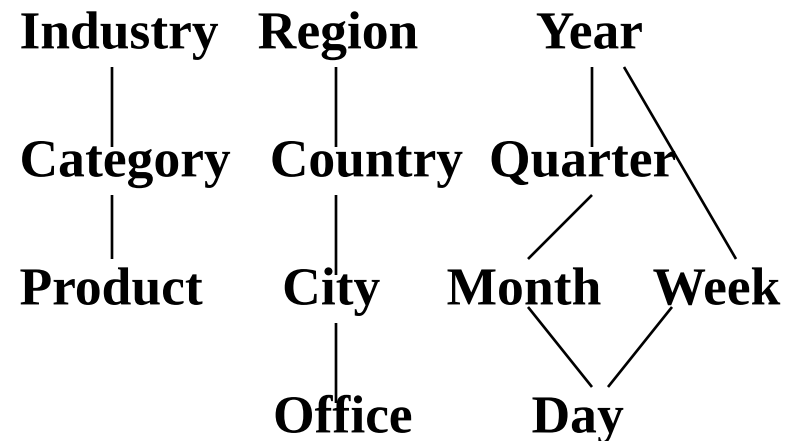
- A **data warehouse** is based on a **multidimensional data model** which views data in the form of **a data cube**, where each side represents a different aspect of the data.
- A data cube, such as **sales**, allows data to be modeled and viewed in multiple dimensions (different perspectives)
 - **Dimension tables**: define the "sides" of the data cube such as **item** (item_name, brand, type), or **time** (day, week, month, quarter, year)
 - **Fact table** contains **measures** (such as **dollars_sold**) and keys (links) to each of the related dimension tables (items, locations)

Multidimensional Data

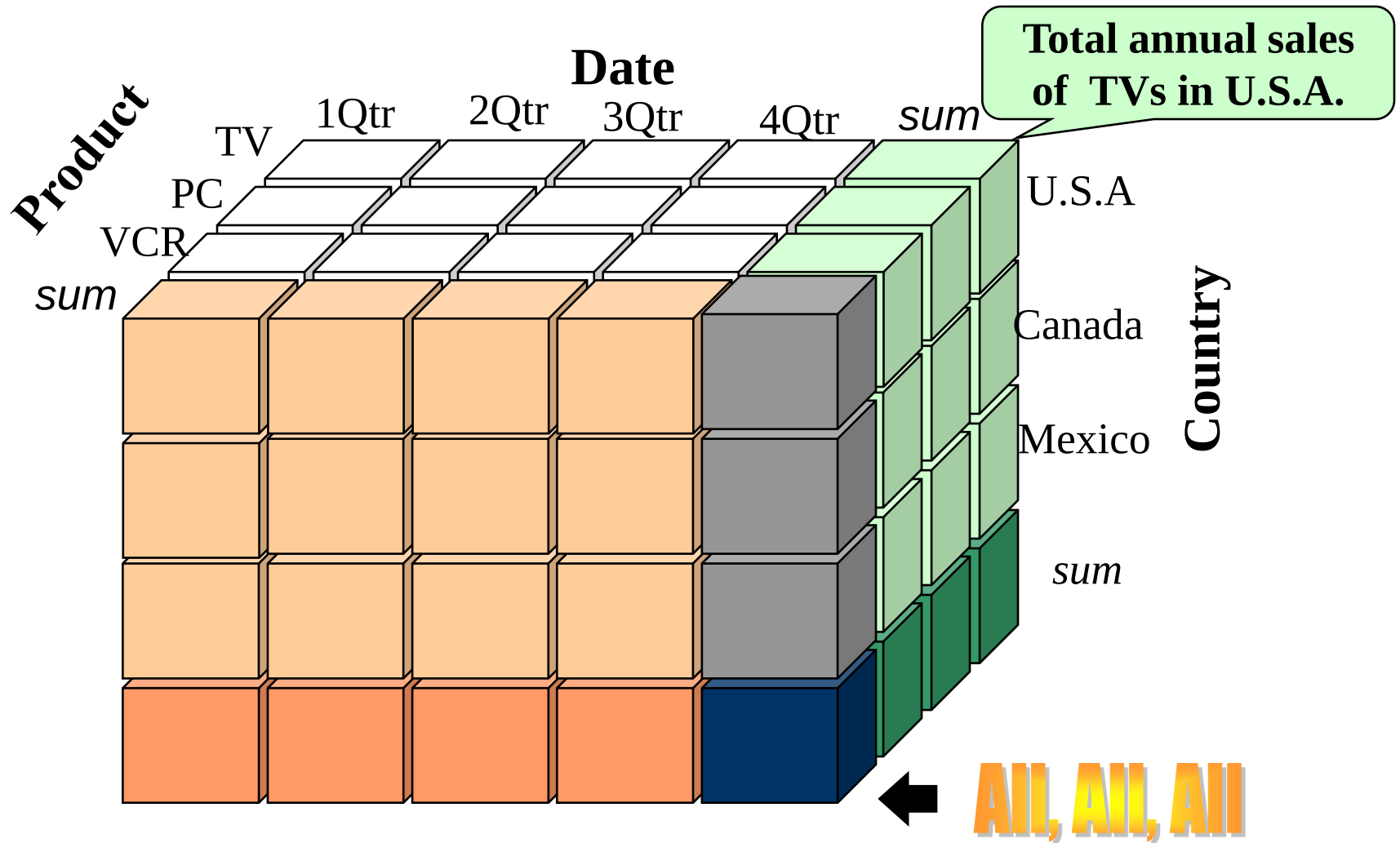
- Sales volume as a function of product, month, and region



Dimensions: *Product, Location, Time*
Hierarchical summarization paths

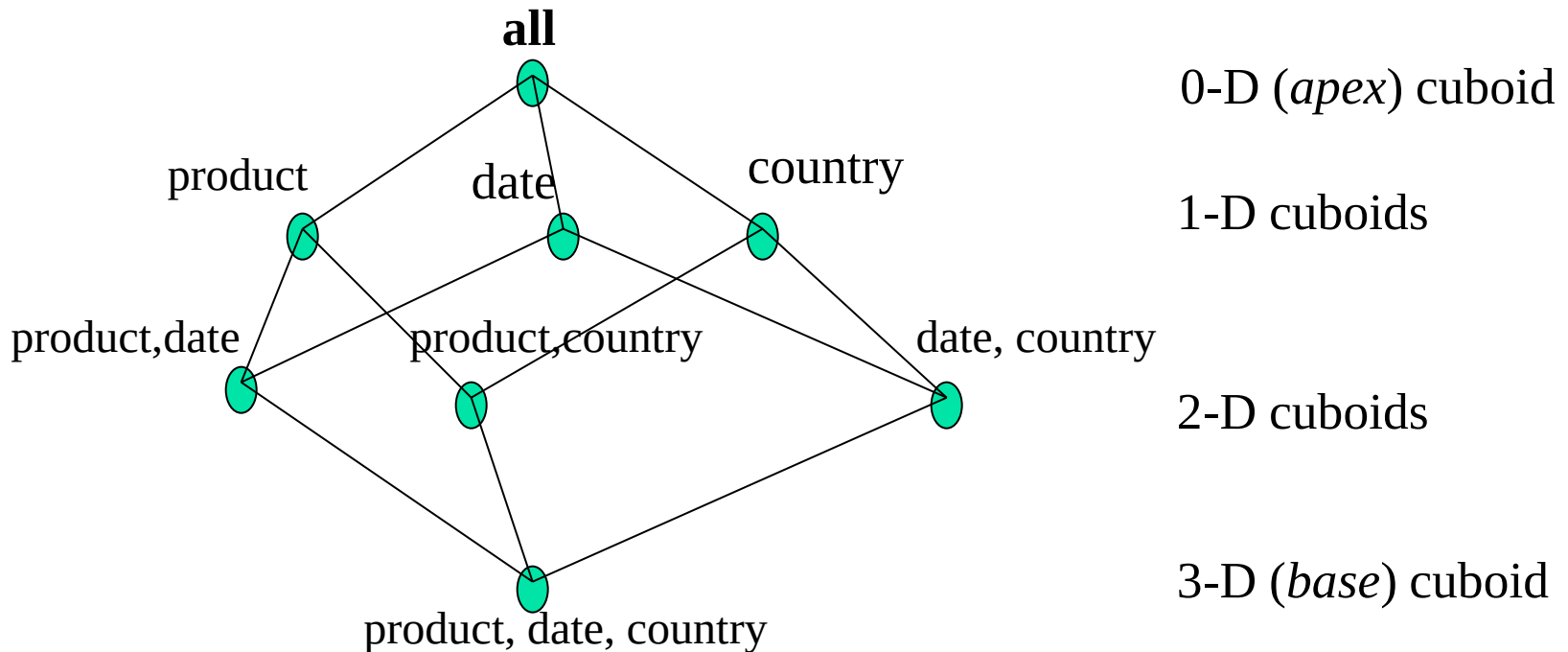


A Sample Data Cube



Cube: A Lattice of Cuboids

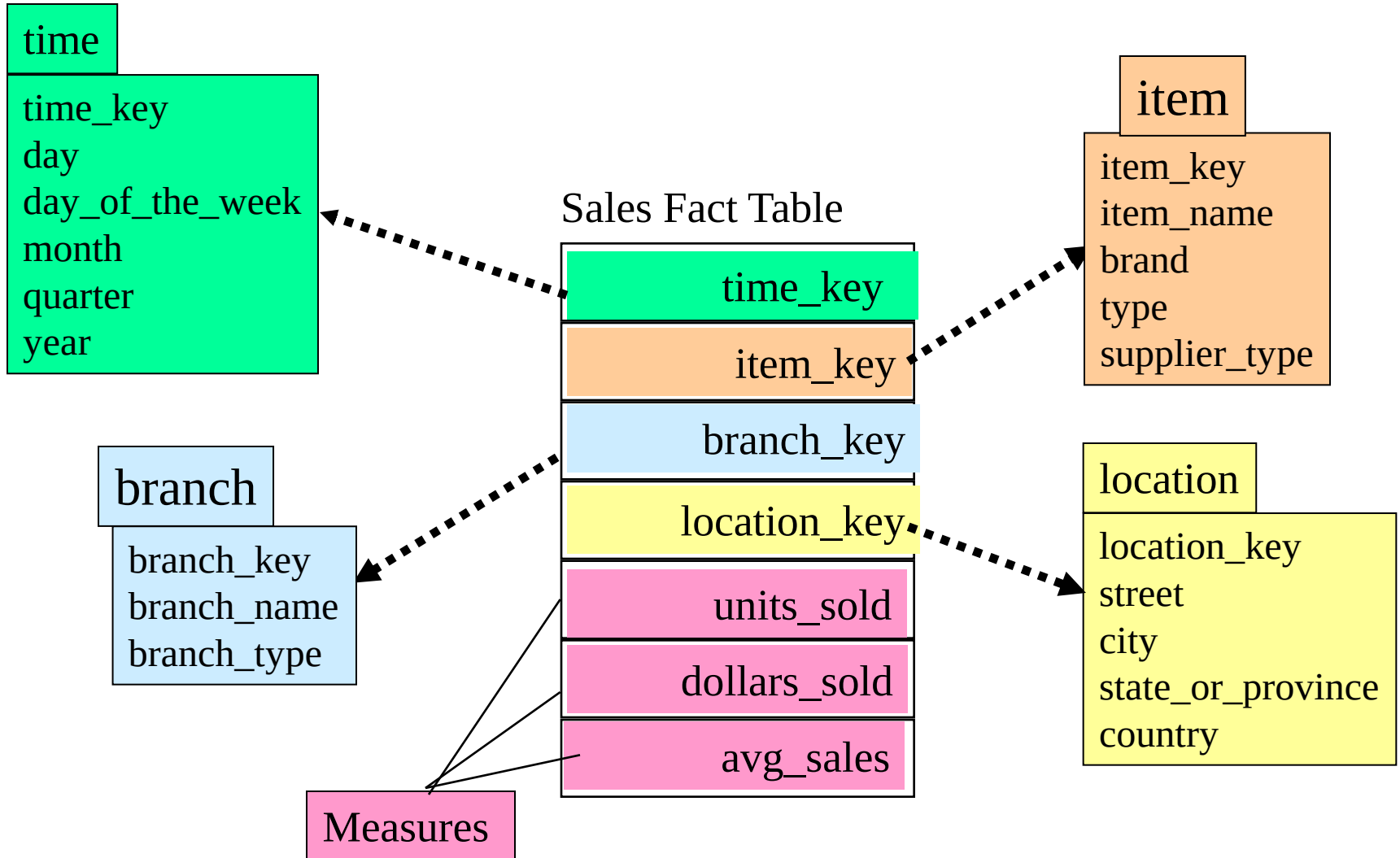
- In data warehousing literature, an n-D base cube is called a **base cuboid** (the most detailed version of data). The top most 0-D cuboid, which holds the highest-level of summarization, is called the **apex cuboid**. The lattice of cuboids (All the possible levels of summarization in between the base and apex) forms a **data cube**.



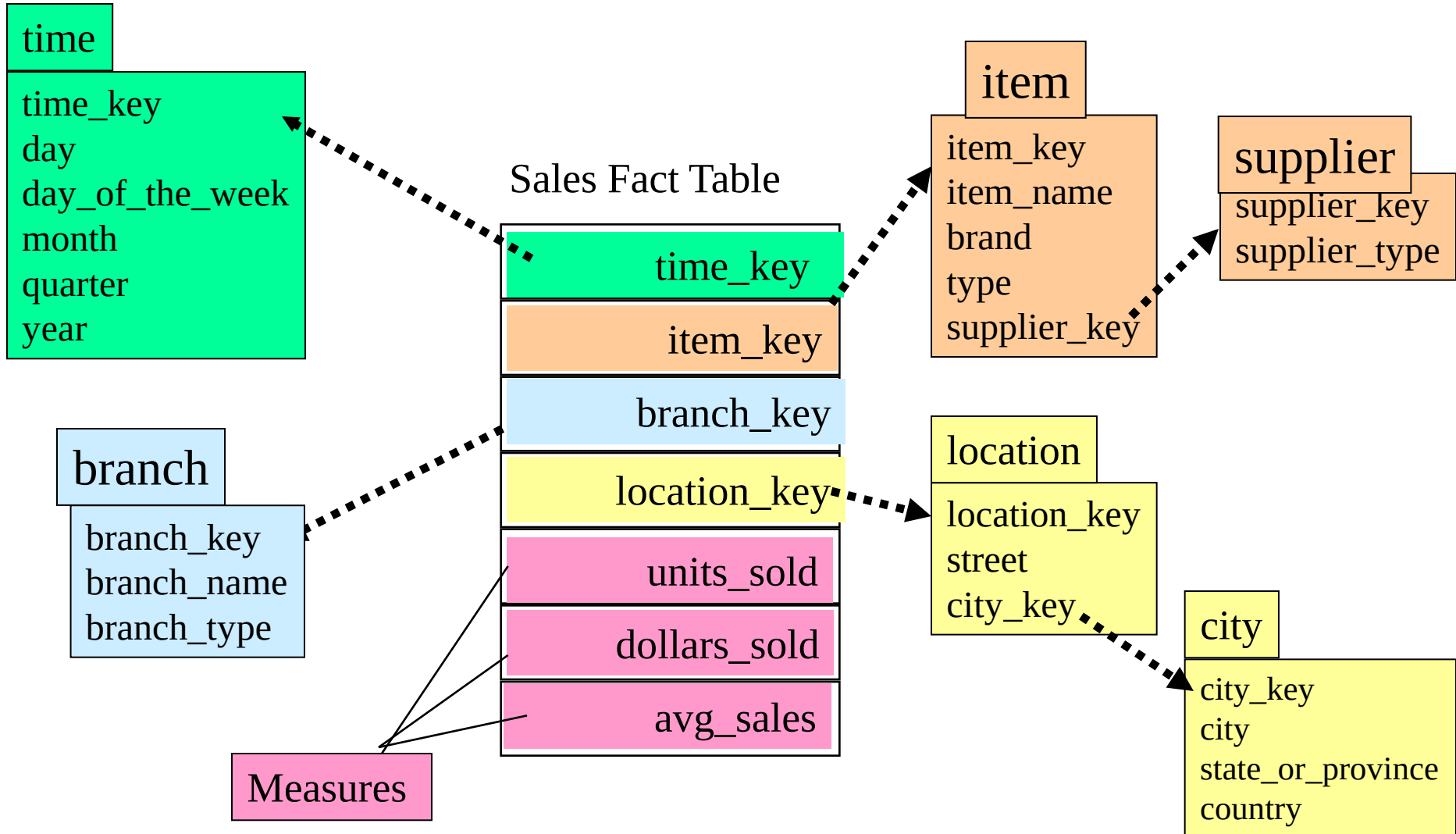
Conceptual Modeling of Data Warehouses

- These are different ways to organize the tables in a data warehouse (dimensions and facts) to make it easier to analyze data:
 - **Star schema**: A fact table in the middle connected to a set of dimension tables
 - **Snowflake schema**: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
 - **Fact constellations**: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

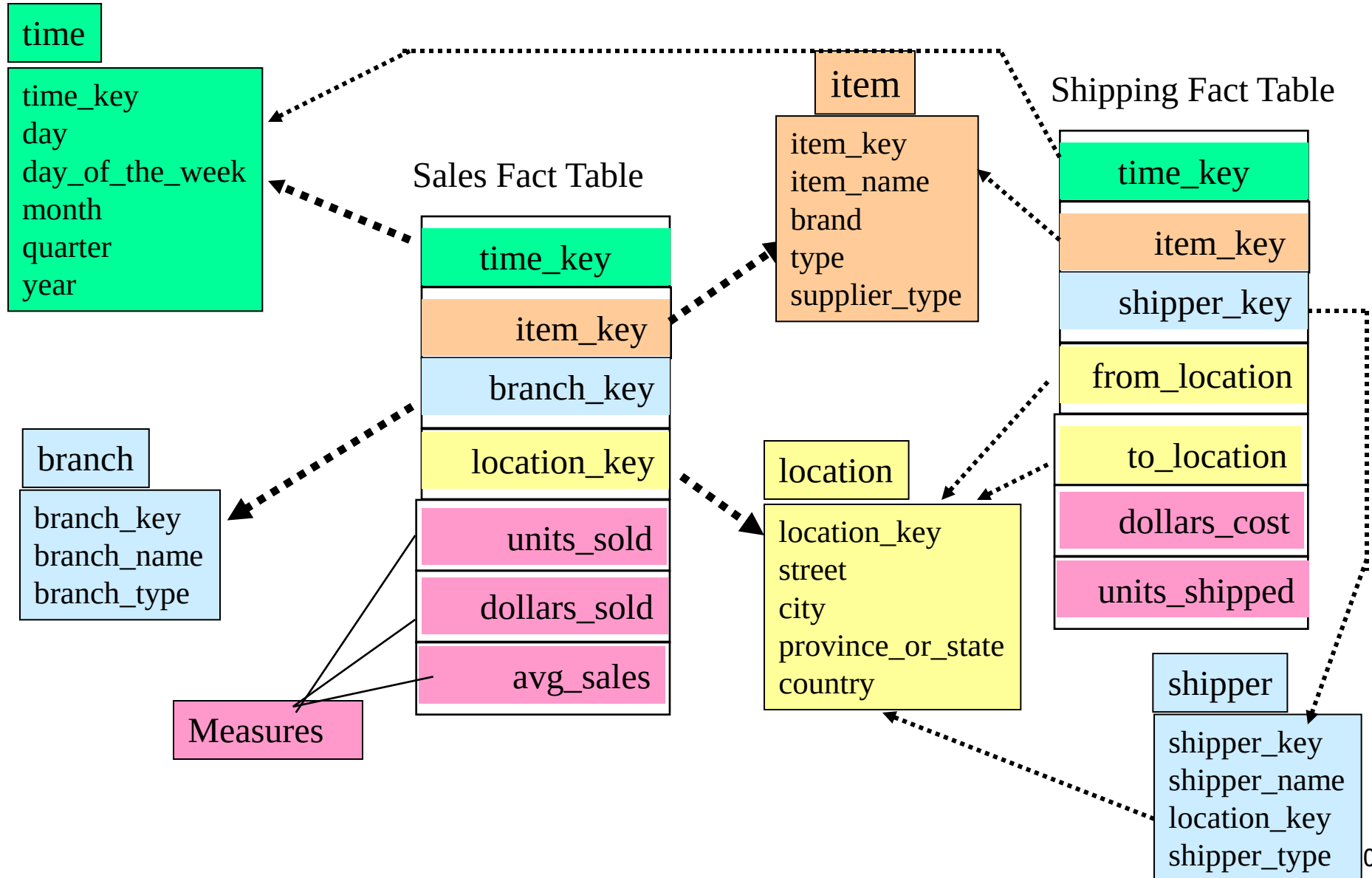
Example of Star Schema



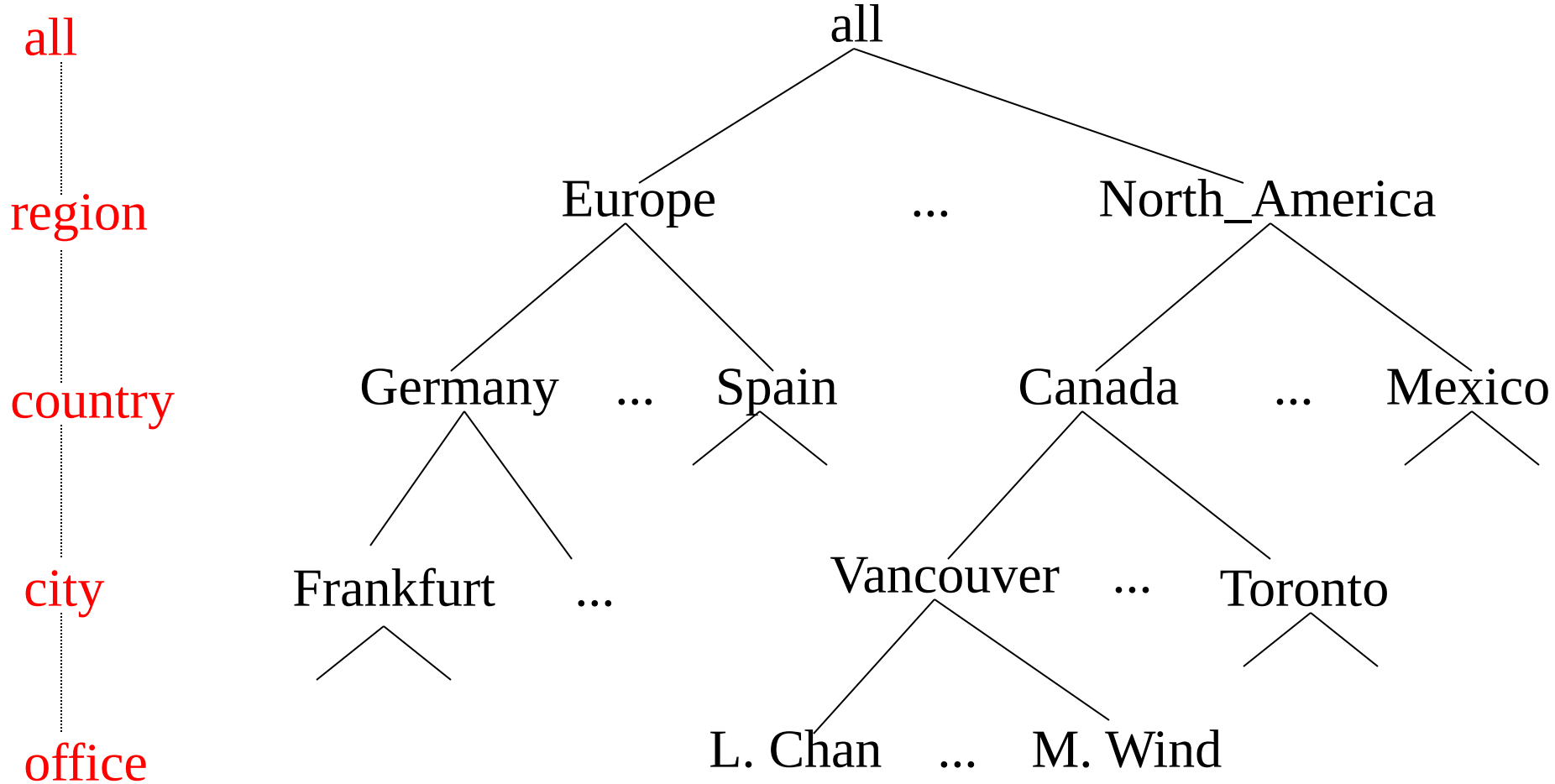
Example of Snowflake Schema



Example of Fact Constellation



A Concept Hierarchy: Dimension (location)



View of Warehouses and Hierarchies

The screenshot displays the dbminer application interface, which is used for managing data warehouses and their dimensions. The main window is titled 'dbminer' and features a menu bar (File, Edit, Query, View, Window, Help) and a toolbar with various icons for file operations and queries.

The interface is divided into several panes:

- Left Pane (Warehouse Hierarchy):** Shows a tree view of the 'DemoWH' warehouse. It includes a 'SCHEMAS' folder containing 'MasterDemoDB.dbo.SalesD', which has 'COLUMNS', 'DIMENSIONS' (with sub-items: Product, Region, revenue, cost, profit, order_qty), 'MEASUREMENTS', 'CUBES' (with sub-items: SalesData_Cube, Small_Cube), and 'DMQLs'. Below this is 'stockdata.dbo.stock' with its own 'COLUMNS', 'DIMENSIONS' (with sub-items: date, price, price1), 'MEASUREMENTS', 'CUBES', and 'DMQLs'.
- Center Pane (Level Name):** Displays a list of dimension levels: 'region', 'country', 'branch_r', and 'rep_name'.
- Right Pane (Warehouse Hierarchy):** Shows a detailed view of the 'ANY' warehouse. It lists geographical regions: 'Europe' (with sub-items: Belgium, France, Germany, Essen, Frankfurt, Spain, Sweden, United Kingdom), 'Far East', 'North America' (with sub-items: Canada, Montreal, Toronto, Vancouver), and 'United States'. The 'Vancouver' region is expanded, showing a list of individuals: Charles Loo Nam, Hari Krain, Kaley Gregson, Lee Chan, Malcom Young, Marthe Whiteduck, and Torey Wandiko.

At the bottom of the window, there is a status bar that reads 'For Help, press F1' and a 'NUM' button.

Typical OLAP Operations

- These operations allow you to explore and analyze data in a data cube from different perspectives.
- **Roll up (drill-up):** summarize data
 - *by moving up hierarchy or by dimension reduction (e.g. From daily sales to monthly sales)*
- **Drill down (roll down):** reverse of roll-up
 - *from higher level summary to lower level summary or detailed data, or introducing new dimensions (e.g. From country sales to city sales)*
- **Slice and dice:** *project and select*
 - *Selecting a specific subset of data by filtering on one or more dimensions.*
 - **Slice:** *Showing sales for "Product A" across all dates and regions. (Fixing one dimension).*
 - **Dice:** *Showing sales for "Product A" in "Region B" during "January 2024." (Fixing multiple dimensions).*

Typical OLAP Operations

- **Pivot (rotate):**

- reorient the cube to change the view or visualization, 3D to series of 2D planes
- *E.g. Swapping rows and columns in a table or Displaying data as a 3D cube and then rotating it to see different face or converting a 3D view into a series of 2D slices.*

- **Other operations:**

- ***drill across:*** involving querying (across) more than one fact table that share common dimensions
- ***drill through:*** through the bottom level of the cube to its back-end relational tables (from a summarized view in the data cube to the detailed data)

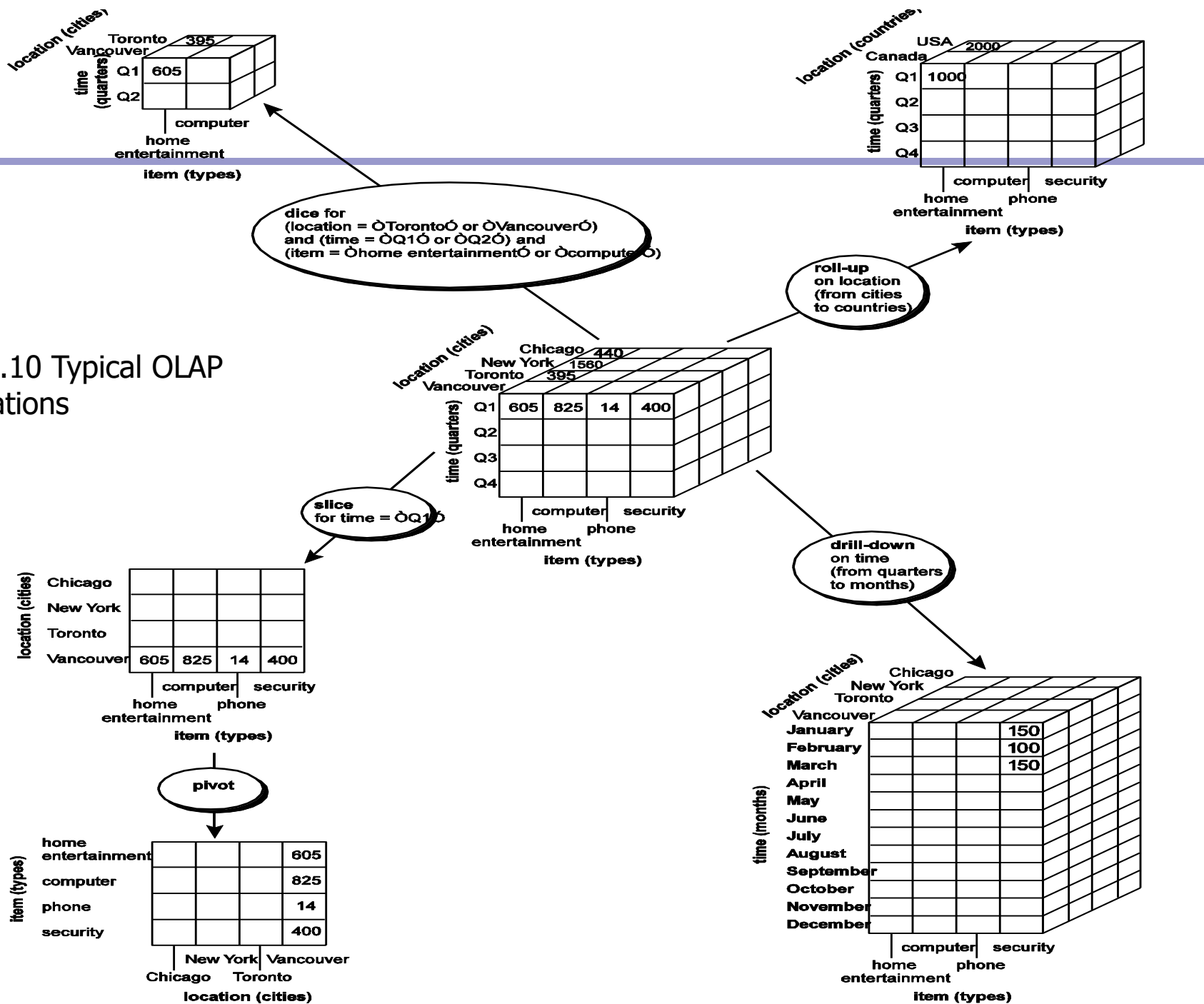


Fig. 3.10 Typical OLAP Operations

Efficient Data Cube Computation

- **Materialization** of data cube refers to how much of that multi-dimensional data is pre-calculated and stored for faster querying.
 - **full materialization:** calculates and stores every possible cuboid. It makes queries very fast but uses a lot of storage.
 - E.g. Pre-calculating total sales for every product, every month, and every region.
 - **No materialization:** calculates cuboids only when needed. This saves storage, but queries can be slow.
 - E.g. Calculating sales totals only when a user requests a report
 - **partial materialization:** calculates and stores only the most frequently used cuboids. This is a balance between speed and storage.
 - E.g. Pre-calculating total sales for the most popular products and regions

Efficient Data Cube Computation

- **Selection of which cuboids to materialize**
 - Based on:
 - **Size:** How much storage space the cuboid needs.
 - **Sharing:** How many users or queries will use the cuboid.
 - **Access Frequency:** How often the cuboid is accessed.

Efficient Processing OLAP Queries

- **Determine which materialized cuboid(s)** should be selected for OLAP operation.
 - Let the query to be processed be on $\{brand, province_or_state\}$ with the condition " $year = 2004$ ", and there are **4 materialized cuboids available**:

1) **cuboid 1:** $\{year, item_name, city\}$

2) **cuboid 2:** $\{year, brand, country\}$

3) **cuboid 3:** $\{year, brand, province_or_state\}$

4) **cuboid 4:** $\{item_name, province_or_state\}$ where $year = 2004$

Which should be selected to process the query?

- **The best choice is Cuboid 3**, it directly matches the dimensions and conditions of your query (brand, province_or_state, year = 2004).
- **Using Cuboid 3:** The system can directly retrieve the data, which will be much faster than calculating it from the base data.

Indexing OLAP Data: Bitmap Index

- **A bitmap index** is a special type of index that uses bit vectors (sequences of 0s and 1s) **to quickly find data**. It's very efficient for certain types of queries, particularly those involving filtering or counting.
- A bitmap index is created for a specific column in a table.
- Each value in the column has a bit vector
- The length of the bit vector: # of records in the base table
- If the i -th row of the original table has the value corresponding to the bit vector, the i -th bit in the vector is set to 1. Otherwise, it's set to 0.
- Bitmap indexes are generally not efficient for columns with many unique values

Indexing OLAP Data: Bitmap Index

- **Example:** the original table with columns "Cust," "Region," and "Type."
 - **Index on Region:** shows three bit vectors, one for each unique region: "Asia," "Europe," and "America."
 - *E.g. For "Asia," the bit vector is 10100. This means that rows 1 and 3 of the base table have "Asia" in the "Region" column*
 - **Index on Type:** shows two bit vectors, one for each unique type: "Retail" and "Dealer."
 - *E.g.: For "Retail," the bit vector is 10010. This means that rows 1 and 4 of the base table have "Retail" in the "Type" column.*

Base table

Cust	Region	Type
C1	Asia	Retail
C2	Europe	Dealer
C3	Asia	Dealer
C4	America	Retail
C5	Europe	Dealer

Index on Region

RecID	Asia	Europe	America
1	1	0	0
2	0	1	0
3	1	0	0
4	0	0	1
5	0	1	0

Index on Type

RecID	Retail	Dealer
1	1	0
2	0	1
3	0	1
4	1	0
5	0	1

Indexing OLAP Data: Bitmap Index

- **Query:** "Find all customers in **Asia** who are **Dealers**."
- **Bitmap Index:**
 - The bit vector for "Asia" is 10100.
 - The bit vector for "Dealer" is 01101.
 - Perform a bitwise AND operation: $10100 \text{ AND } 01101 = 00100$.
- **Result:** The resulting bit vector 00100 indicates that the third row (C3) satisfies the query.

Base table

Cust	Region	Type
C1	Asia	Retail
C2	Europe	Dealer
C3	Asia	Dealer
C4	America	Retail
C5	Europe	Dealer

Index on Region

RecID	Asia	Europe	America
1	1	0	0
2	0	1	0
3	1	0	0
4	0	0	1
5	0	1	0

Index on Type

RecID	Retail	Dealer
1	1	0
2	0	1
3	0	1
4	1	0
5	0	1

OLAP Server Architectures

- These are different ways to store and access data for multidimensional analysis:
 - **Relational OLAP (ROLAP):** Use relational or **extended-relational DBMS** to store and manage warehouse data and OLAP middle ware (Greater scalability)
 - **Multidimensional OLAP (MOLAP):** Sparse **array-based multidimensional** storage engine
 - Fast indexing to pre-computed summarized data
 - **Hybrid OLAP (HOLAP)** (e.g., Microsoft SQLServer): Combines the best of **ROLAP and MOLAP**
 - Flexibility, e.g., low level: relational, high-level: array
 - **Specialized SQL servers (e.g., Redbricks)**
 - Specialized support for SQL queries over star/snowflake schemas

Design of Data Warehouse: A Business Analysis Framework

- Four views regarding the design of a data warehouse, ensuring it meets the needs of various stakeholders:
 - **Top-down view**
 - focuses on the overall business objectives, allows selection of the relevant information necessary for the data warehouse
 - **Data source view**
 - exposes the information being captured, stored, and managed by operational systems
 - **Data warehouse view**
 - consists of fact tables and dimension tables
 - **Business query view**
 - sees the perspectives of data in the warehouse from the view of end-user (how end-users will interact with the data warehouse to answer their business questions)

Data Warehouse Design Process

- **Typical data warehouse design process**
 - Choose a **business process** to model, e.g., orders, invoices, etc.
 - Select a specific area of the business to focus on
 - Choose the ***grain (atomic level of data)*** of the business process
 - Decide the level of detail for the data.
 - Choose the **dimensions** that will apply to each fact table record
 - Identify the categories used to analyze the data
 - Choose the **measure** that will populate each fact table record
 - Select the numeric values to be analyzed

Data Warehouse Usage

- Three kinds of data warehouse applications

1) Information processing:

- the most basic use of a data warehouse. It's about getting straightforward answers to specific questions and presenting the data in a clear way.
- supports querying, basic statistical analysis, and reporting using crosstabs, tables, charts and graphs
- E.g. A marketing analyst wants to create a chart showing the number of website visitors each day.

Data Warehouse Usage

- Three kinds of data warehouse applications

2) Analytical processing:

- multidimensional analysis of data warehouse data to find patterns and insights.
- supports basic OLAP operations, slice-dice, drilling, pivoting
- E.g. A product manager wants to see how sales of a particular product have changed over time in different regions.

Data Warehouse Usage

- Three kinds of data warehouse applications

3) Data mining

- knowledge discovery from hidden patterns and relationships in the data
- Users use algorithms and techniques to find associations, build models, and make predictions.
- supports associations, constructing analytical models, performing classification and prediction, and presenting the mining results using visualization tools
- E.g. A bank wants to build a model to predict which customers are likely to default on their loans

From On-Line Analytical Processing (OLAP) to On-Line Analytical Mining (OLAM)

- Why **online analytical mining** (combining data mining with OLAP)?
 - **High quality of data in data warehouses**
 - Data warehouse contains integrated, consistent, cleaned data, **,making it reliable for mining.**
 - **Available information processing** structure surrounding data warehouses
 - OLAP tools to explore the data before mining
 - **On-line selection of data mining functions**
 - Integration and swapping of multiple mining functions, algorithms, and tasks to find the best solution for your problem.

References

- Business Intelligence: Data Mining and Optimization for Decision Making. Carlo Vercellis, John Wiley & Sons.
- "The Data Warehouse Toolkit" by Ralph Kimball and Margy Ross
- Data Mining: Concepts and Techniques, Jiawei Han, Micheline Kamber and Jian Pei